



Participant Debriefing Sheet

Study Title: Mitigating Implicit Bias in Clinical Decision Support Systems through Fatigue detection

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1. What is the purpose of this study?

In high-pressure medical environments, clinicians often face fatigue. This state can lead to a reliance on errors which may inadvertently introduce implicit biases into patient triage.

This study tests an **Adaptive User Interface (AUI)** designed to act as a guardian. The system uses machine learning and real-time behavioral tracking to detect when a user might be drifting into a biased decision-making state and provides nudges to help them return to an analytical, evidence-based approach.

2. What will I be asked to do?

If you agree to participate, the study consists of three main stages:

1. **Implicit Association Test (IAT):** A brief computer-based task to measure automatic social associations. This score helps the system personalize its assistance to you.
2. **Training:** You will be taught a set of objective clinical triage rules.
3. **Triage Simulation:** You will categorize several patient profiles while under a simulated time pressure. During this phase, the interface may intervene with alerts or visual cues if it detects signs of cognitive conflict or fatigue.

3. What data is being collected?

To make the interface Aware, the system collects:

- **Behavioral Biomarkers:** High-frequency mouse movement coordinates and timing.
 - **Mouse Tortuosity (τ):** The system calculates the "straightness" of your mouse path. A jagged path ($\tau > 1$) is an objective indicator of internal hesitation or conflict between a biased shortcut and a clinical rule.

4. How is my data protected?

- **Anonymization:** Your name is never linked to your data. You will be assigned a **Unique Subject ID**.
- **Aggregate Analysis:** We publish results only as group trends. For example: *"The system reduced errors by X% across all participants."*
- **Secure Storage:** Data is stored on encrypted University of York drives.

5. What are the risks and benefits?

- **Risks:** You may experience minor mental fatigue or temporary stress due to the time-pressured nature of the task.
- **Benefits:** You will gain insight into how AI and machine learning can be used to improve human decision-making and safety in professional settings.

6. Is participation voluntary?

Yes. You can stop the study at any time without giving a reason.

Contact Information

If you have questions about the logic used in this study or how your data is handled, please contact:

- **Researcher:** ed1304@york.ac.uk
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